

Independent Axial Seamount Thermomechanical Model: Progress Report

Axial Modeling Project

October 2026

Abstract

This report records an independent PyLith implementation of the written thermomechanical method for Axial Seamount, with observational checking against authorized Ocean Observatories Initiative (OOI) records and original raw BPR channels from earlier Axial deployments. It uses no paper-produced observations, corrections, numerical results, model output, or figure data. The project verifies elastic and one-branch Maxwell responses, steady three-dimensional heat conduction, temperature-property transfer, and postprocessed failure connectivity, and synthetic three-branch historical forward checks. Central and Eastern OOI histories provide 3,927 common finite daily records from 2014 through 2026; the aggregate quality code is retained as NOT EVALUATED. Original raw NCEI BPR channels add deployments from 1987 through 2002, including the 1998 event, and raw MGDS channels add records through 2013, including the 2011 event. The one-branch OOI diagnostic fits pressure through static elastic compliance, then produces a large Central displacement error and an early modeled failure path. The historical three-branch event checks use synthetic branch parameters and have held-out South biases of 0.329 m and 0.371 m. Additional deployment checks span 1995–2013; the 2011–13 South comparison has 1.242 m RMSE and -1.214 m bias. These outputs remain provisional: the ellipsoid compliance is not mesh-converged, generalized Maxwell branch parameters are unavailable, Eq. 16 is inconsistent with its stated modulus limits, and the paper’s written thermal equation specifies zero heat production without mechanical feedback. The current work therefore supports a partial implementation record, not a reproduction of the published eruption forecasts.

1 Scope and data provenance

The project reconstructs model inputs from written descriptions of the study and its supplement, then implements independent solvers and checks. The main model article and publisher-served supplement are specification sources [1, 2]. No observation series, corrections, numerical results, model output, source code, plotting scripts, or figure data produced for the paper are used. The report does not digitize plotted curves or use published numerical results. The supplement copy carries a confidential manuscript footer, so extracted parameters remain version-qualified.

OOI daily BPR depth records cover Central and Eastern Caldera. The products report relative uplift after detiding and drift correction; their aggregate quality flag is NOT EVALUATED [4]. The analysis retains flag code 2 without filtering. Common finite observations span 2014-09-05 through 2026-09-30. Raw NCEI records provide uncorrected pressure from ten Axial deployments in 1987–96 and further deployments in 1997–99 and 2000–02. WC09–WC32 use 56.25-second samples; WC51 onward uses 15-second samples. Selected MGDS files provide original `Depth` or `RawDep` channels from Center and South deployments between 2003 and 2013 [5, 6]. The workflow excludes derived detided, filtered, and drift-corrected fields; source files and processed daily series remain outside version control. The

MGDS archive uses the CC BY-NC-SA 3.0 license, and its data citation and license are preserved in the repository data notes.

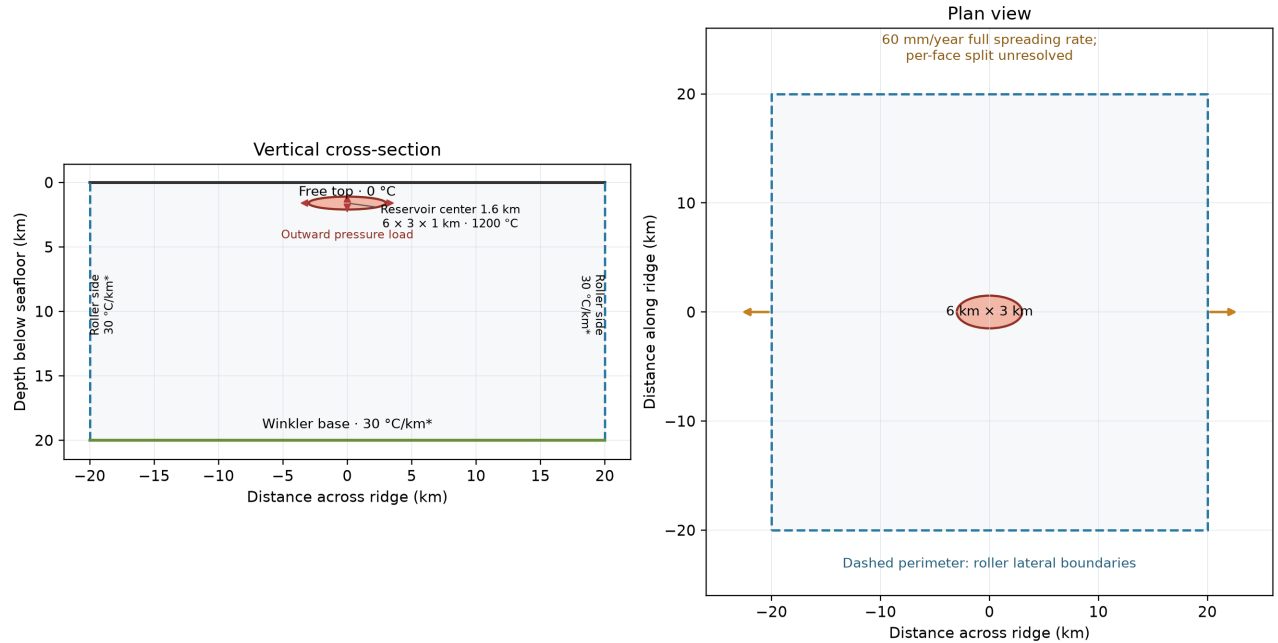
2 Written model and implementation

The specified reservoir is an ellipsoid measuring 6 by 3 by 1 km, centered 1.6 km below the seafloor. The written thermal method is steady conduction,

$$\nabla \cdot (k \nabla T) = -Q, \quad Q = 0, \quad (1)$$

with a 30 °C/km background geotherm, a 0 °C surface, and a 1200 °C reservoir boundary. The hydrothermal conductivity law is implemented as a separate case. Because lateral and basal thermal conditions are not specified, the current finite-element calculation extends the background geotherm to all outer faces and records this as an assumption.

Axial thermomechanical model setup — schematic



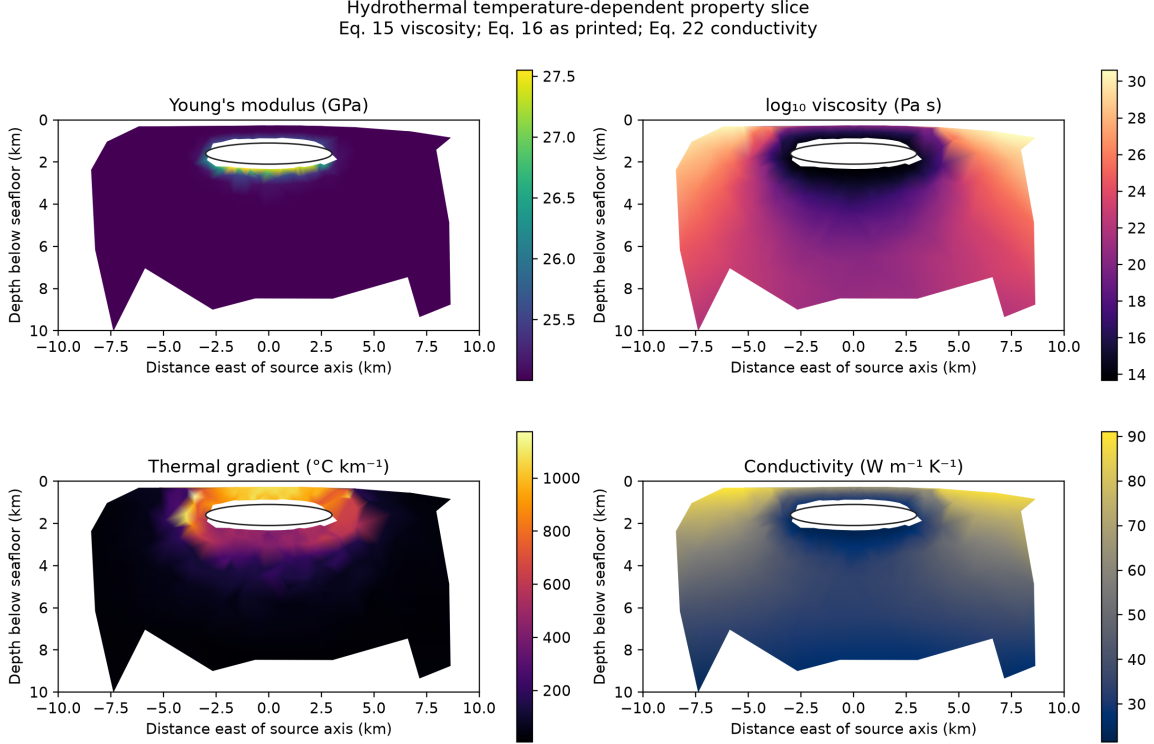
The 40 × 40 × 20 km box is a project fallback; the written model extent is unavailable. *The side and basal 30 °C/km geotherm is an explicit thermal assumption. Reservoir depth is 1.6 km to center; Winkler stiffness and the tectonic face-rate split are unresolved.

Figure 1: Schematic of the written reservoir geometry and boundary conditions. The 40 by 40 by 20 km model box is a project fallback; the ellipsoid dimensions and center depth follow the written specification. Roller sides, a Winkler base, a free top, and the 60 mm/year full spreading rate are labeled without assigning an unresolved per-face velocity. The 30 °C/km side and basal thermal conditions are project assumptions. Generated by `make model-setup-schematic`; no published figure or geological data were used.

The Arrhenius viscosity law is evaluated from the temperature field. The printed temperature-dependent Young’s modulus equation increases from about 25 GPa toward 75 GPa as temperature rises, contrary to the written brittle and ductile descriptions. The implementation preserves this equation only as a diagnostic; it does not claim the trend is physically resolved. The written Maxwell model requires a generalized branch spectrum that is not fully specified. Current PyLith runs therefore use one Maxwell

branch with explicitly assumed properties. The PyLith 5.0.2 material interface supplies properties and initial state through an auxiliary spatial database; a same-mesh restart can transfer state when properties change [3].

The hydrothermal thermal solve now produces an exploratory midplane map of temperature-dependent properties (Fig. 2). This slice uses the written Arrhenius viscosity and conductivity relations and evaluates Eq. 16 exactly as printed. The finite-thickness slice, outer-face geotherm, and incomplete generalized Maxwell spectrum limit comparison with the complete four-rheology panels.



Cell-center projection within $|y| \leq 0.7$ km; white ellipse marks the reservoir. Side and basal temperatures use the assumed $30^\circ\text{C}/\text{km}$ geotherm. Eq. 16 conflicts with the stated brittle and ductile limits.

Figure 2: Hydrothermal temperature-dependent properties on a finite-thickness slice around the model $y=0$ plane. Cell-center values show Young’s modulus from Eq. 16 as printed, Arrhenius viscosity, the finite-element temperature gradient, and Eq. 22 conductivity. The white ellipse marks the reservoir. Side and basal temperatures extend the assumed $30^\circ\text{C}/\text{km}$ geotherm. Generated by `make thermal-property-slices`; Eq. 16 remains an unresolved transcription diagnostic.

The project has verified a three-dimensional steady heat solve, a synthetic thermal-to-property-to-mechanics transfer, and same-mesh Maxwell restart. An unchanged-property segmented run differs from a continuous run by at most 1.612×10^{-8} in displacement, stress, total strain, and viscous strain. A separate uniform 1200°C material swap carries viscous strain across the restart boundary and changes the final displacement by 50.08 percent relative to the uniform-property run. This verifies the restart interface; the artificial uniform-temperature swap is not a coupled physical case.

A manufactured check transfers an affine temperature field from a six-tetrahedron source mesh to the 2,761-tetrahedron mechanics mesh, with a maximum interpolation error of $1.14 \times 10^{-13}^\circ\text{C}$. The smoke case also solves the written hydrothermal model on a distinct 3,060-tetrahedron ellipsoid mesh. Sampling at mechanics-cell centers yields finite temperatures from 8.769 to $1,066.240^\circ\text{C}$. PyLith completes a two-second Maxwell solve using the mapped heterogeneous material database and writes finite stress and viscous strain; peak stress is 1.75811×10^7 Pa. The smoke case verifies point sampling of a physical

steady field and material-database use across meshes. It does not test conservative transfer, time-varying properties, or two-way thermal-mechanical feedback.

The paper’s thermal method specifies $Q = 0$ and gives no stress- or deformation-generated heat term. The roadmap’s mechanics-to-thermal return coupling therefore lacks a written physical law. The current one-way OOI diagnostic does not add an invented heat source. This prevents a complete two-way reproduction until the return coupling is specified or approved as a separate exploratory extension.

3 Independent observations and model checks

Figure 3 shows the available Central and Eastern OOI uplift records. Each instrument uses its own first-sample reference. The plot is an observation-only partial record: it has no raw pre-2014 values, eruption markers, or earthquake counts.

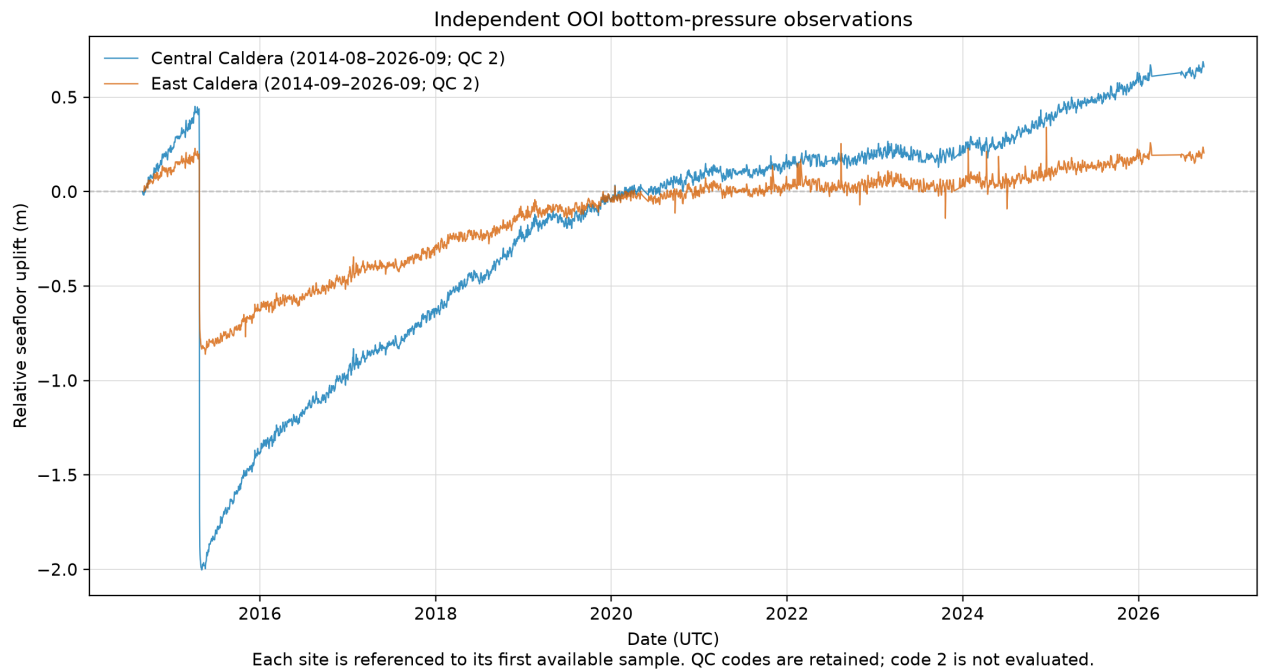


Figure 3: Daily OOI relative uplift at Central and Eastern Caldera from each site’s first available record. Generated by `make bpr-observation-plot`; aggregate quality code 2 (NOT EVALUATED) is retained. The data show the observed interval available for independent model checking.

The original raw NCEI channels add intermittent Axial deployment records from 1987 through 2002, while MGDS raw channels extend coverage through 2013. The deployment-context figure places every deployment on its own first-sample baseline; gaps between deployments are not interpolated, and no tide, ocean, or sensor-drift correction is applied. WC09–WC32 were sampled every 56.25 seconds; later NCEI and MGDS records use 15-second samples.

An independent Fox archive from MGDS (IEDA/322344; doi:10.1594/IEDA/322344) for the same 1997–98 Center and South instruments agrees with NCEI relative uplift over 309 Center and 365 South days, with correlations above 0.999999 and RMSE values of 0.050 and 0.016 m. This checks raw-channel handling across archives; it does not add independent station coverage. The archive’s detided and low-pass-filtered fields are excluded.

An elastic Mogi point source is calibrated to Central uplift and checked against the Eastern record. The

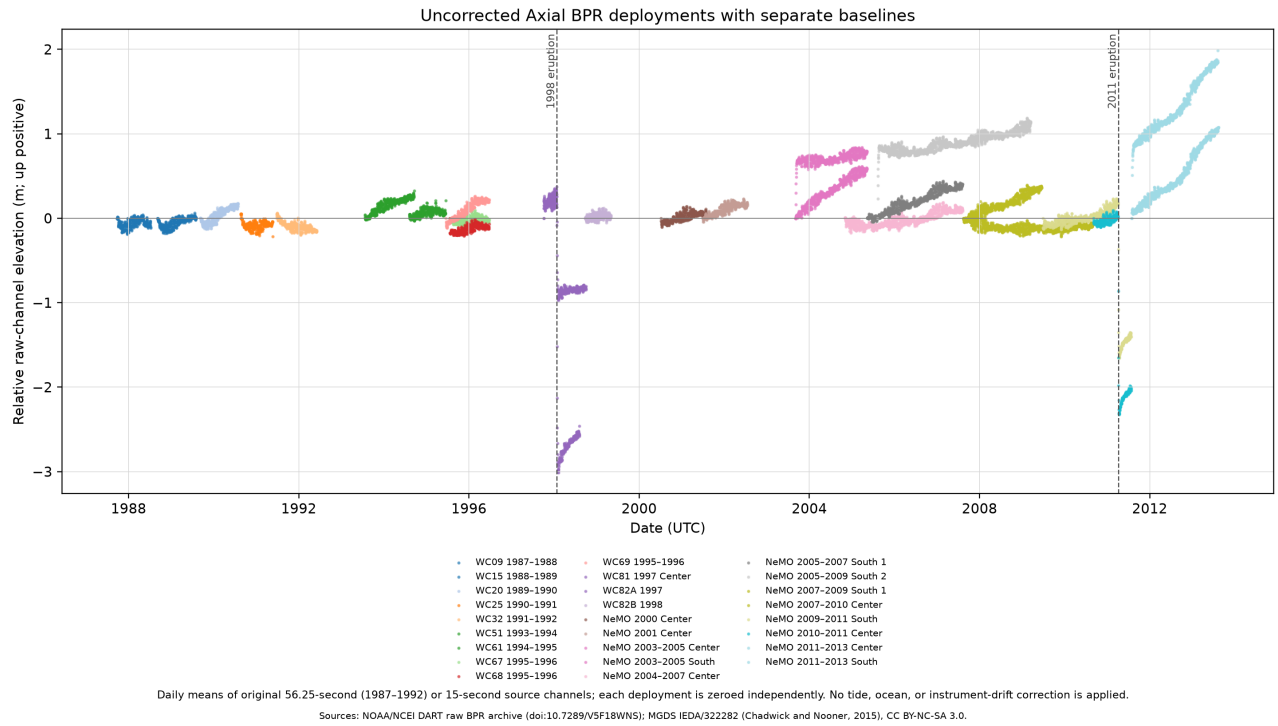


Figure 4: Separate-baseline daily means of original raw non-OOI BPR channels from 1987 through 2013. The plot shows intermittent deployment coverage; each segment is zeroed independently and remains uncorrected for tides, ocean variability, and instrument drift. Generated by `make bpr-historical-check`. Sources: NOAA/NCEI DART raw BPR archive (doi:10.7289/V5F18WNS) and MGDS IEDA/322282 (Chadwick and Nooner, 2015). MGDS-derived values are distributed under CC BY-NC-SA 3.0.

inferred pressure spans -2.96 to $+1.06$ GPa; Eastern RMSE is 0.0594 m with 0.995 correlation. The spatial correlation does not validate this pressure scale. It shows that the point-source proxy cannot supply the intended historical ellipsoid pressure model.

The raw historical channels add Center-to-South checks at the 1998 and 2011 eruptions, a 1995–96 pre-eruption overlap, and three later deployment overlaps. The 1995–96 WC68/WC69 overlap spans 338 paired daily values. Its held-out South prediction has 0.180 m RMSE, $+0.159$ m bias, and 0.876 correlation; the raw record’s slow South trend is not captured by the static model. The later pairs span 614 paired days in 2003–05, 572 in 2007–09, and 731 in 2011–13. Static Mogi predictions have South RMSE values of 0.134 , 0.156 , and 0.387 m, with respective biases of $+0.113$, $+0.136$, and $+0.363$ m. These channels retain tides, oceanographic variability, and sensor drift, and the fits omit viscoelastic memory. They extend observation checking across the pre-OOI interval; they are not calibrated pressure histories or eruption forecasts.

The three-branch historical forward checks use first-common-day zeroing and infer each daily pressure from the static Center compliance. In the 1998 pair, the Center and held-out South RMSE values are 0.186 m and 0.503 m; the 2011 values are 0.114 m and 0.680 m. South biases are $+0.329$ m and $+0.371$ m, respectively. The checks use synthetic branch reference viscosities and fractions, retain raw ocean and instrument effects, and inherit the nonconverged static compliance. They test the historical loading path without calibrating rheology or producing a hindcast.

The saved stress histories for all seven raw-BPR-driven windows are also evaluated with the current Mohr–Coulomb proxy: cohesion is 1 MPa, friction angle is 25 degrees used directly as ϕ , and pore pressure is zero. A tensile cutoff is not applied to the shear path. The resulting cavity-to-surface paths appear within the first 196 days of all seven windows, including the five intervals without an eruption. In 1998 a path is already present at the first saved output, so onset is only bounded above by seven days. In 2011 linear stress interpolation brackets onset at 17.61 days after the first common BPR day. The same proxy therefore fails to distinguish the eruption windows from inter-eruption intervals and is not calibrated to eruption timing.

BPR window	First path time	Saved records with a path
1995–96	≤ 7 days	46/49
1998	≤ 7 days	44/44
2003–05	25.38 days	83/88
2005–07	195.84 days	71/117
2007–09	67.86 days	68/83
2011	17.61 days	30/47
2011–13	≤ 7 days	105/106

Table 1: Provisional Mohr–Coulomb cavity-to-surface path checks on the historical three-branch stress histories. Time is measured from each independently zeroed BPR window. Thresholds use 1 MPa cohesion, 25 -degree friction angle, and zero pore pressure, without a tensile cutoff. Onsets after the first saved record are interpolated linearly between saved PyLith stress records; paths already present in the first output are bounded by seven days.

Five additional raw Center/South overlaps extend the three-branch check across 1995–2013. Table 2 summarizes the held-out South comparisons. Each window has its own first-day zero, and deployment gaps are left unfilled. The 2011–13 correlation of 0.991 coexists with a 1.242 m RMSE and -1.214 m bias. Large residuals, changing correlations, synthetic branch parameters, and uncorrected records prevent these plots from establishing a historical calibration.

The same model runs also predict two additional raw BPR locations that were not used to define their primary Center/South pairs. WC67 is held out from the 1995–96 WC68-driven run over 338

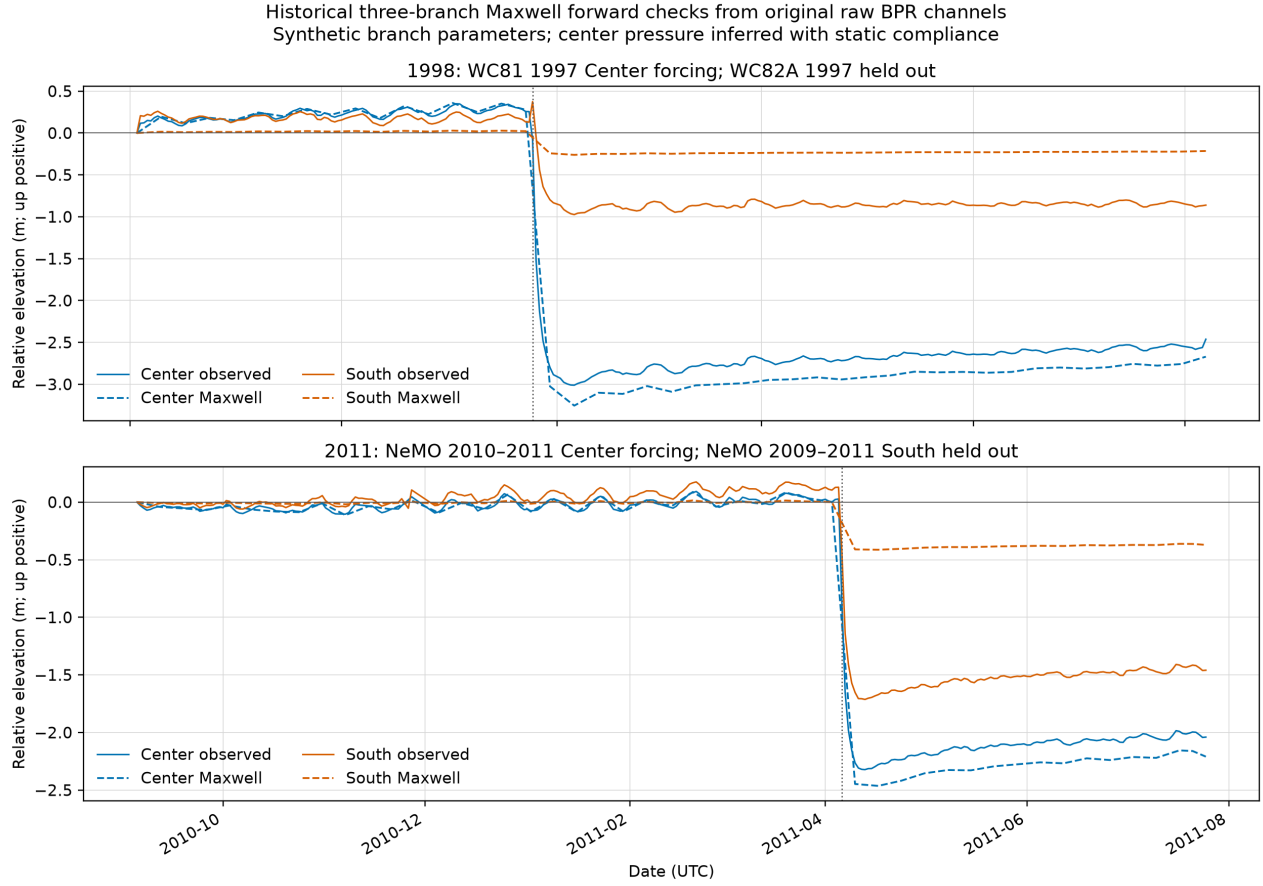


Figure 5: Three-branch generalized Maxwell forward checks against raw daily BPR observations for the 1998 and 2011 Center/South deployment pairs. Center pressure is inferred from static elastic compliance; South is held out. Synthetic branch properties, first-common-day zeroing, uncorrected raw records, and nonconverged compliance limit interpretation. Generated by `make historical-generalized-maxwell-check`.

Interval	RMSE (m)	Bias (m)	Correlation
1995–96	0.183	−0.161	0.793
2003–05	0.651	−0.649	0.660
2005–07	0.157	−0.135	0.958
2007–09	0.123	−0.101	−0.355
2011–13	1.242	−1.214	0.991

Table 2: Held-out South residuals for five raw deployment overlaps.

days; NeMO South 1 is held out from the 2007–09 Center-driven run over the 572 days shared with the primary window. The WC67 residual is small, while the second South record has larger error and negative correlation. The contrasting station results show that spatial transfer varies across deployments and that each raw channel remains a separate model check.

Held-out raw station	Window	RMSE (m)	Bias (m)	Correlation
WC67	1995–96	0.036	−0.021	0.685
NeMO South 1	2007–09	0.221	−0.191	−0.506

Table 3: Additional independent BPR station checks using the existing Center-driven Maxwell solutions. Neither station contributes to the pressure history used in its model run.

The elastic ellipsoid comparison improves the pressure scale but remains mesh-sensitive. On the 2,761-tetrahedron mesh, unit-load compliance is 0.0318994 m/MPa at Central and 0.00345580 m/MPa at Eastern. The static Central-calibrated pressure spans −62.21 to 22.16 MPa. Eastern RMSE is 0.221 m, relative L2 error is 76.36 percent, and correlation is 0.995. Across tested refinements, station compliance changes by 21–76 percent; these results do not establish a stable pressure calibration. The bounded station-region comparison uses 2,761, 3,031, 3,124, 3,053, and 3,325 tetrahedra. These independently generated unstructured meshes are not guaranteed to be nested: the 950 m case has fewer elements than the 1,000 m case and changes Central and Eastern compliance by 59.8 and 58.1 percent. The 900 m case then changes them by −35.1 and −31.5 percent. Repeating the five solves reproduced their results exactly, but the sequence has no compliance plateau and no adjacent local-refinement pair meets the 5 percent criterion at both stations.

The one-branch Maxwell diagnostic uses monthly pressure inferred from the static elastic Central compliance, linearly interpolated between samples. Its 147 stress records cover 12.07 years. Central RMSE is 1.096 m (correlation 0.787); Eastern RMSE is 0.195 m (correlation 0.922). The Mohr–Coulomb post-processor uses $C = 1$ MPa, applies the tabulated 25 degrees directly as the friction angle, assumes zero pore pressure, and applies no tensile cutoff because tensile strength is unavailable. It finds a cavity-to-surface path in 146 records, first at a saved time of 60 days. These are diagnostic outcomes, not eruption predictions.

The static pressure conversion does not account for Maxwell memory. A second OOI-only check instead fits Central uplift directly through a PyLith one-branch Maxwell response kernel and reserves Eastern for a spatial holdout. For this assumed rheology, the Central RMSE is 0.00495 m and Eastern RMSE is 0.225 m (correlation 0.983). The inferred pressure spans −60.4 to 11.8 MPa. The direct PyLith history agrees with kernel superposition to relative L2 errors of 0.00028 at Central and 0.00080 at Eastern. This verifies the linear response implementation; it does not validate the pressure scale, which still depends on a nonconverged mesh, GCV smoothing, and an assumed single Maxwell branch.

4 Numerical verification

The bounded PyLith 5.0.2 toolchain runs an elastic-cavity smoke case on 2,761 linear tetrahedra. Analytical Mogi checks verify positive inflation uplift, radial symmetry, and pressure scaling. A PyLith spherical-cavity comparison has a 33.2 percent nearest-axis error on 3,191 tetrahedra; this coarse check does not establish mesh convergence. Increasing the half-width and bottom depth from 8 km to 12 km reduces peak uplift by 61.4 percent and raises fixed-grid vector error from 40.4 to 58.0 percent on independently generated, nonnested meshes. This comparison does not isolate the finite-boundary effect; neither mesh validates the analytical response quantitatively. The tetrahedral thermal solver matches manufactured linear and uniform-source solutions. Its full ellipsoid thermal case converges numerically,

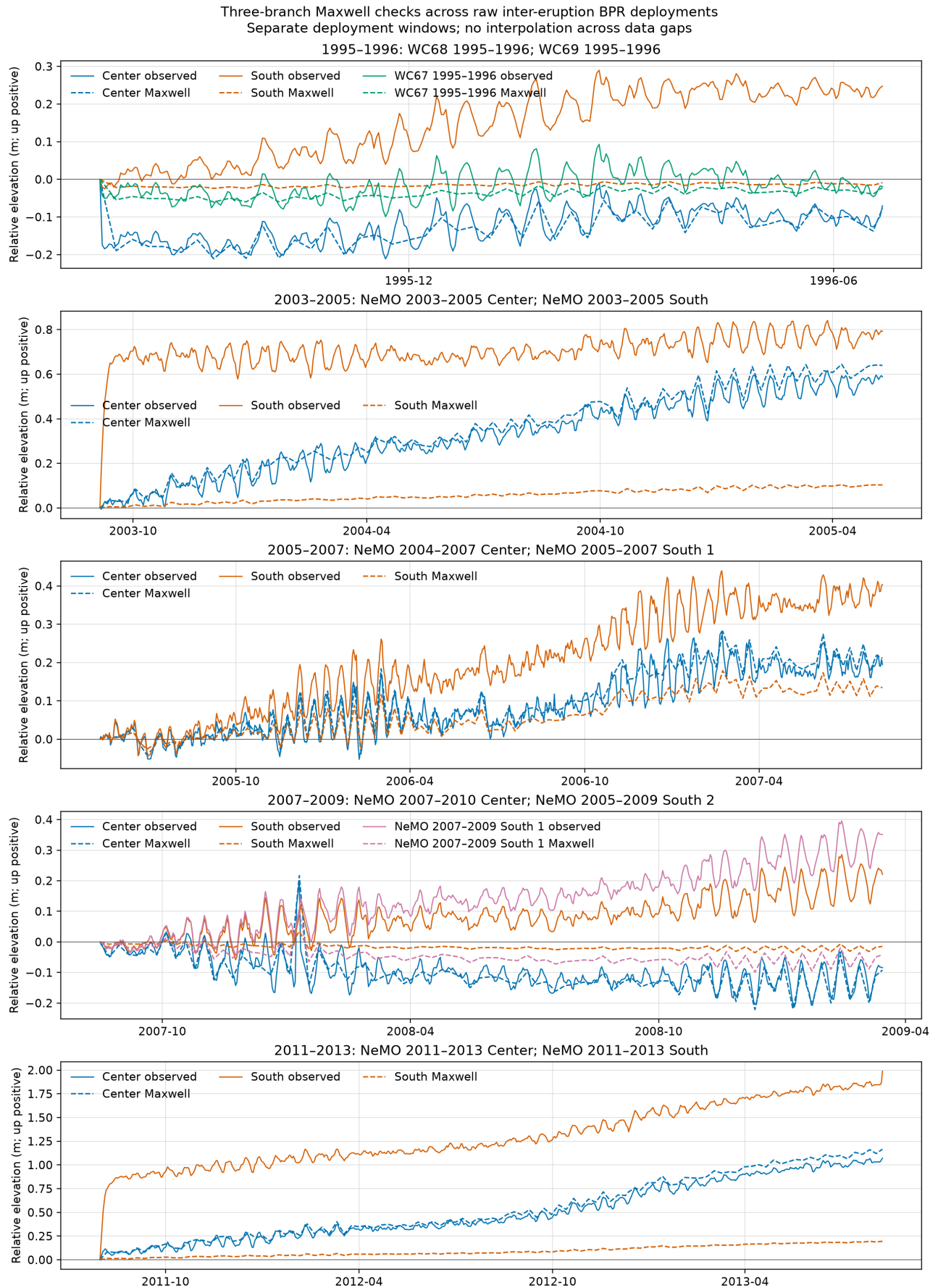


Figure 6: Three-branch Maxwell checks for five raw Center/South deployment overlaps from 1995 through 2013, with WC67 added to the 1995–96 panel and NeMO South 1 added to the 2007–09 panel as held-out raw BPR stations. Each deployment pair is independently zeroed; gaps are not interpolated. Center pressure uses static elastic compliance, and the synthetic branch parameters and uncorrected

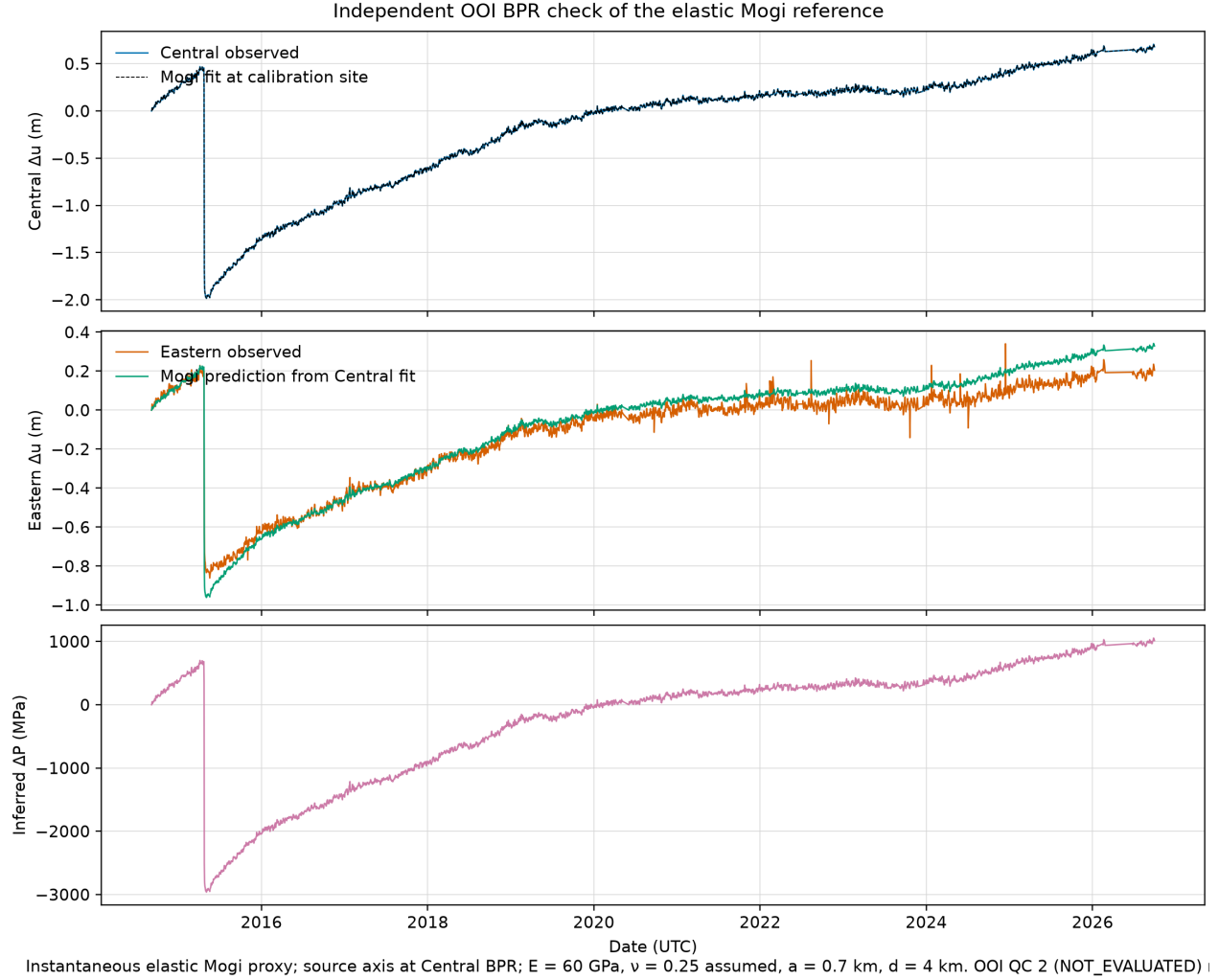


Figure 7: OOI observations and elastic Mogi point-source response. Generated by `make bpr-mogi-check` from independent OOI records; Central is fit by construction and Eastern is held out. The inferred pressure scale is physically unsuitable for the later reservoir model.

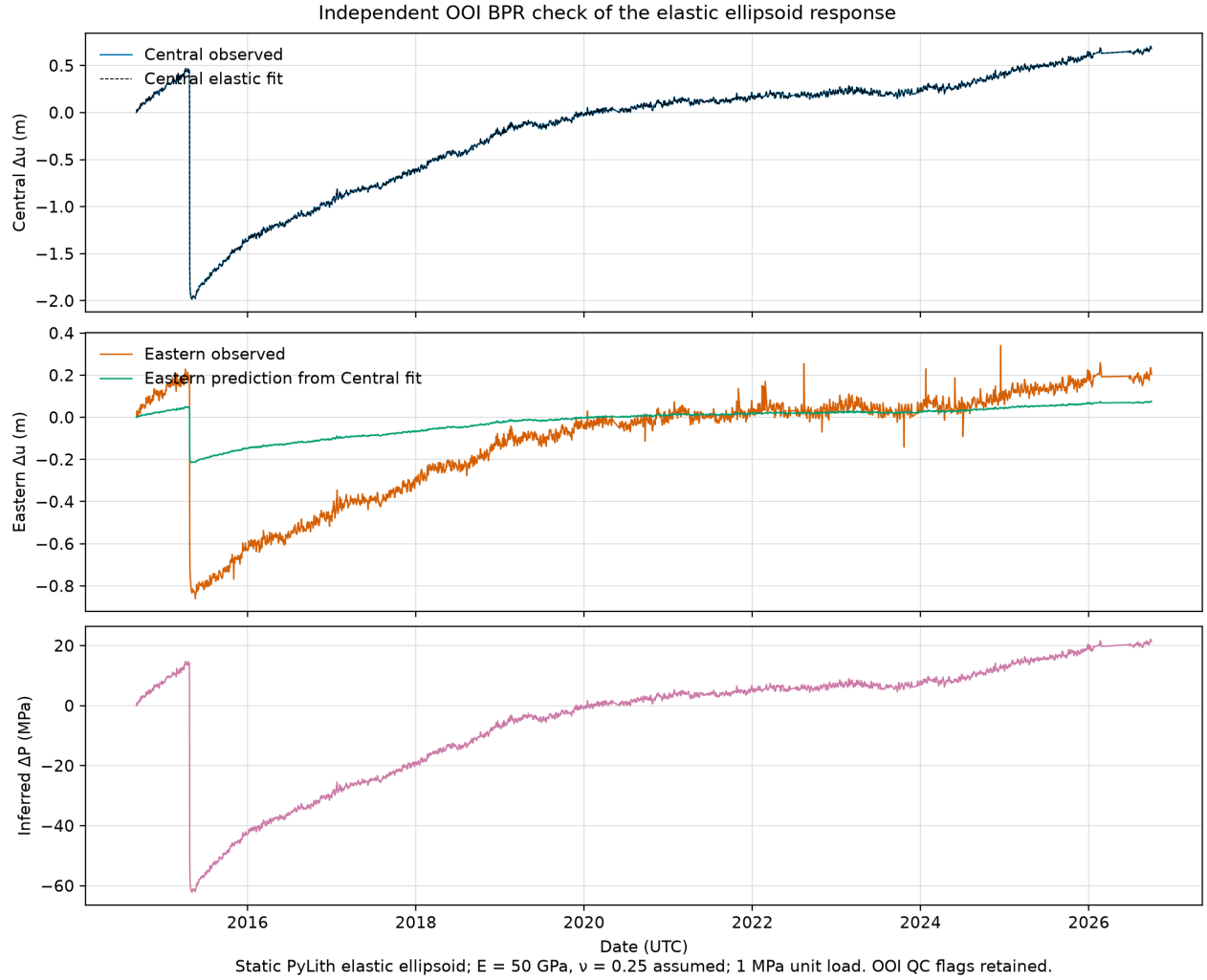
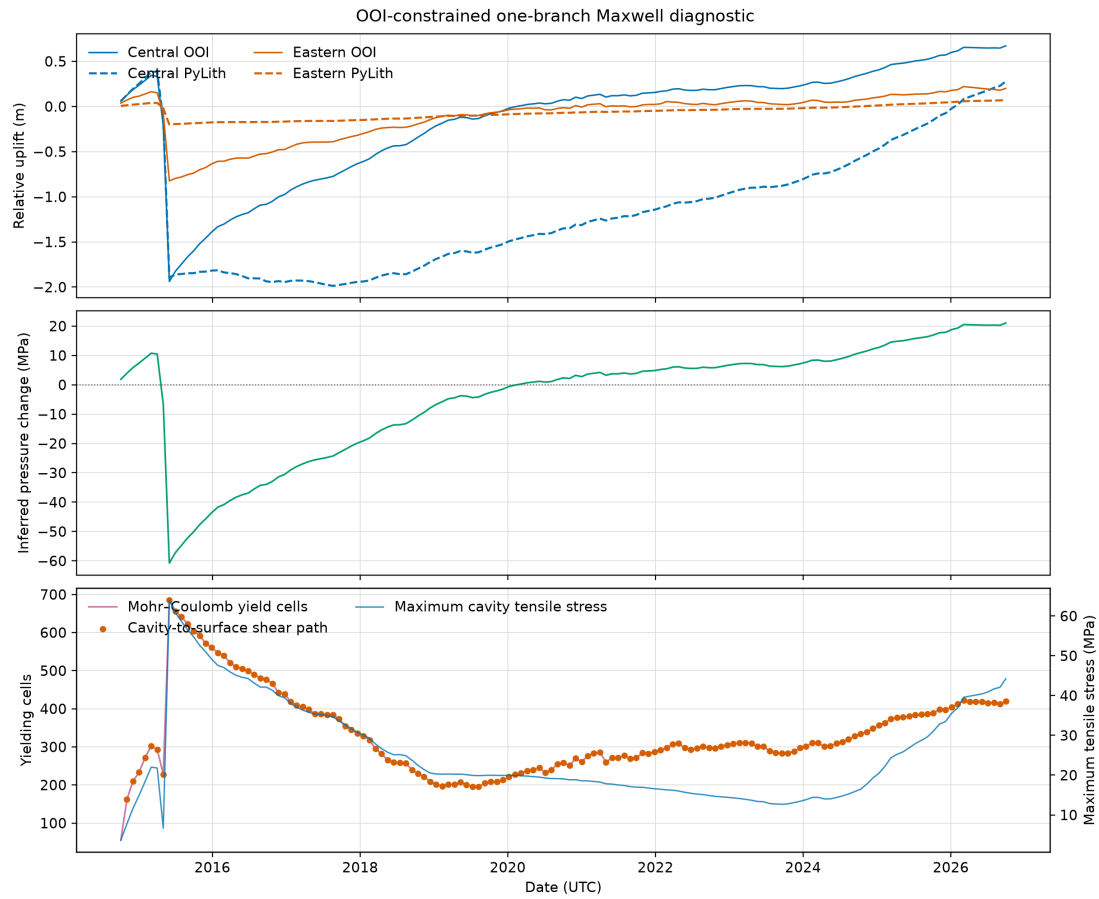
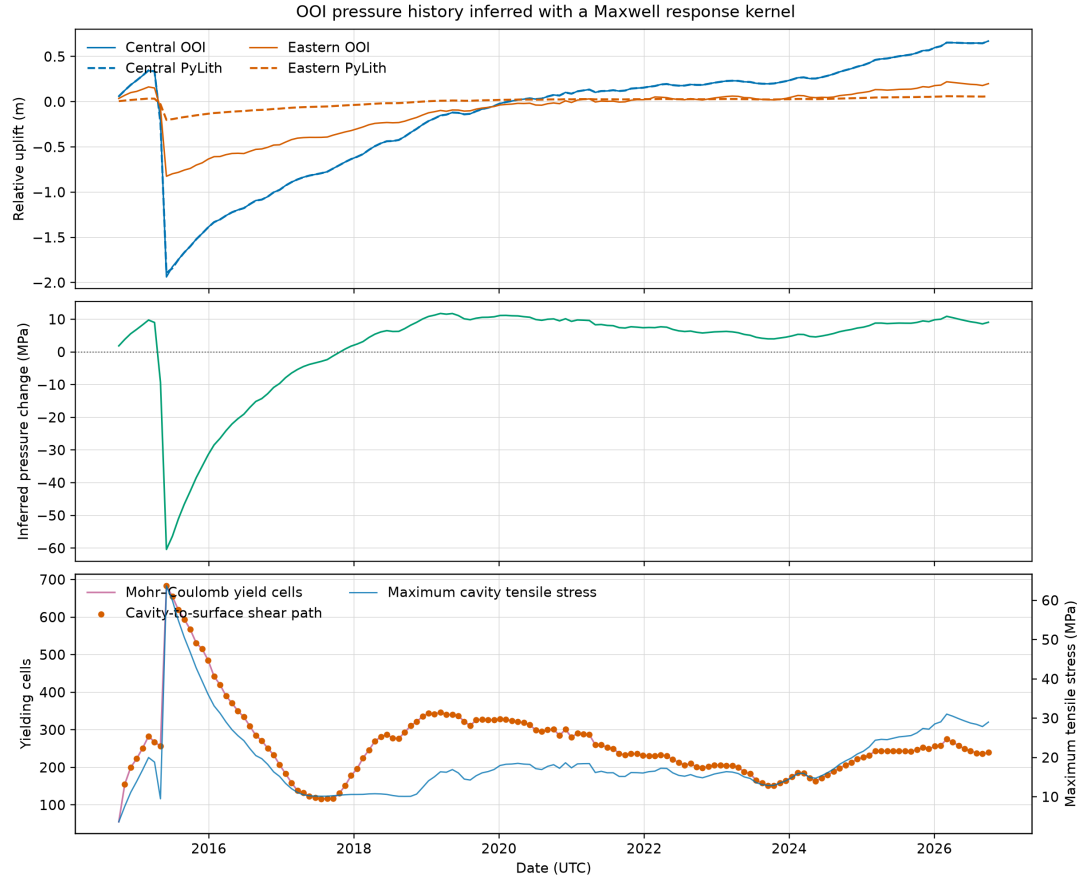


Figure 8: Static elastic ellipsoid fit to Central OOI uplift and Eastern holdout prediction. Generated by `make ellipsoid-bpr-check`. The amplitude mismatch and nonconverged compliance make the inferred pressure provisional.



OOI QC 2 (NOT_EVALUATED) retained. Pressure uses static elastic Central compliance. Failure uses $C = 1$ MPa, 25° directly, zero pore pressure, and no tensile cutoff. The 2,761-tetrahedron compliance is not mesh-converged.

Figure 9: OOI-constrained one-branch Maxwell diagnostic. The panels compare OOI and modeled uplift, static-fit pressure, and stress-derived failure indicators. Generated by `make ooi-maxwell-history-plot`. Quality code 2 is retained; pressure uses static compliance; the mesh and failure assumptions remain provisional.



Central-only pressure inversion; Eastern held out. Single uniform Maxwell branch, GCV second-difference smoothing, and coarse ellipsoid mesh. OOI QC 2 (NOT_EVALUATED) retained. Failure proxy is provisional; no tensile cutoff.

Figure 10: OOI Central pressure inversion with a one-branch Maxwell response kernel. Central is fitted and Eastern is held out; dashed curves show the direct PyLith history. The plot includes the inferred pressure and provisional failure diagnostics. Generated by `make ooi-maxwell-pressure-inversion` from independent OOI records. The smoothness prior, pressure scale, and nonconverged mesh remain provisional.

but its lateral and basal boundary temperatures remain assumptions.

The failure analysis converts PyLith Cauchy stress to principal stresses, applies a Mohr–Coulomb yield test, and searches face-connected tetrahedra between the cavity and surface. Synthetic tests verify state ordering, boundary identification, and time-resolved path detection. Those checks validate software behavior under stated assumptions; they do not validate a physical failure threshold.

5 Panel-by-panel status

No manuscript numerical panel is independently reproduced. Figure 3 is an observation-only partial record; Figures 7–10 are project-generated diagnostics. The detailed command, output, and comparison inventory is maintained in `docs/figure_reproduction.md`.

Panel	Status and current limitation
Fig. 1a	Not reproduced; required bathymetry, lava-flow, and event source records are excluded.
Fig. 1b	Schematic only; the fallback box and unresolved boundary conventions preclude a numerical panel comparison.
Fig. 2	Partially reproduced from OOI records for 2014–2026 and intermittent raw BPR deployments for 1987–2013; generalized Maxwell checks span six separate raw deployment windows. The continuous history, eruption markers, and event counts are absent.
Fig. 3, elastic	Not reproduced; the required property slices and full written configuration are unavailable.
Fig. 3, viscoelastic	Not reproduced; a one-branch smoke test does not provide generalized rheology or panel fields.
Fig. 3, temperature-dependent	Not reproduced; Eq. 16 is inconsistent with its stated limits and the fields are incomplete.
Fig. 3, hydrothermal	Partially reproduced; one hydrothermal property slice is available, but the full rheology panels and resolved Eq. 16 are not.
Fig. 4a	Not reproduced; observations span 1997–2026 with gaps, and the full pressure history and mesh-converged compliance remain unavailable.
Fig. 4b	Not reproduced; the full calibrated reservoir-volume and historical deformation record is unavailable.
Fig. 5a	Not reproduced; event-scale stress history and rheology mapping are unavailable.
Fig. 5b	Not reproduced; event-scale stress history and rheology mapping are unavailable.
Fig. 5c	Not reproduced; event-scale stress history and rheology mapping are unavailable.
Fig. 5d	Not reproduced; event-scale stress history and rheology mapping are unavailable.
Fig. 5e	Not reproduced; event-scale stress history and rheology mapping are unavailable.
Fig. 5f	Not reproduced; event-scale stress history and rheology mapping are unavailable.
Fig. 5g	Not reproduced; event-scale stress history and rheology mapping are unavailable.
Fig. 5h	Not reproduced; event-scale stress history and rheology mapping are unavailable.
Fig. 5i	Not reproduced; event-scale stress history and rheology mapping are unavailable.
Fig. 5j	Not reproduced; event-scale stress history and rheology mapping are unavailable.
Fig. 5k	Not reproduced; event-scale stress history and rheology mapping are unavailable.
Fig. 5l	Not reproduced; event-scale stress history and rheology mapping are unavailable.
Fig. 5m	Not reproduced; event-scale stress history and rheology mapping are unavailable.
Fig. 5n	Not reproduced; event-scale stress history and rheology mapping are unavailable.
Fig. 5o	Not reproduced; event-scale stress history and rheology mapping are unavailable.
Fig. 5p	Not reproduced; event-scale stress history and rheology mapping are unavailable.
Fig. 5q	Not reproduced; event-scale stress history and rheology mapping are unavailable.
Fig. 5r	Not reproduced; event-scale stress history and rheology mapping are unavailable.
Fig. 5s	Not reproduced; event-scale stress history and rheology mapping are unavailable.

Panel	Status and current limitation
Fig. 5t	Not reproduced; event-scale stress history and rheology mapping are unavailable.
Fig. 6a–d	Not reproduced; no conceptual redraw has been made.
Supp. S1	Partially reproduced for one hydrothermal temperature-dependent case; complete slices and all four rheologies are not available.
Supp. S2	Not reproduced; the hydrothermal brittle–ductile transition comparison is incomplete.
Supp. S3	Not reproduced; independent Del Negro and prior FEM benchmarks are incomplete.
Supp. S4	Not reproduced; a verified Winkler-foundation implementation is unavailable.
Supp. S5	Schematic only; side and basal geotherm, fallback domain, and tectonic face-rate split remain qualified assumptions.
Supp. S6	Not reproduced; mesh-converged OOI pressure calibration and the full rheology set are unavailable.

6 Limitations and next work

The present implementation is a collection of verified components, OOI checks, and raw historical BPR diagnostics. It does not implement the four rheology configurations, a generalized Maxwell spectrum, a verified Winkler foundation, or the complete historical pressure calibration. The model-box dimensions, Poisson ratio, host density, tensile strength, and several source conventions remain unresolved. The finite-element compliance is not mesh-converged.

The written thermal equation is steady and sets heat production to zero, while the project roadmap asks for mechanics-to-thermal feedback. No return-coupling law is specified in the written model. This report preserves that gap instead of silently inventing a source term. A scientifically faithful reproduction cannot claim full two-way coupling until the governing feedback and missing parameters are specified. Raw historical BPR checks add eruption and inter-eruption observations, but their uncorrected channels and static elasticity do not reproduce the 1998–2011 stress evolution.

Static PyLith ellipsoid checks across five raw overlap windows from 1995–2013 yield seven held-out station RMSE values: 0.180/0.028 m (WC69/WC67, 1995–96), 0.257 m (2003–05), 0.166 m (2005–07), 0.124/0.244 m (South 2/South 1, 2007–09), and 0.614 m (2011–13). The 2011–13 comparison has +0.551 m bias despite 0.993 correlation; NeMO South 1 is anticorrelated at -0.325 . Raw ocean and instrument variability and the unconverged mesh limit interpretation.

References

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6. Chadwick, W., et al. (2023), Processed Bottom Pressure Recorder data from uncabled instruments deployed at Axial Seamount, Marine Geoscience Data System, doi:<https://doi.org/10.1594/IEDA/322282>. The analysis reads only original raw **Depth** and **RawDep** channels and excludes the accompanying corrections.